Assignment - 1

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Your Roll number: 25027 $\quad p=3 \quad q=3 \quad 91=1 \quad K=6$

1. $a=3$, $b=6$ $\rightarrow$ $f(x) = 3x^4 - 4(a+b)x^3 + 6ab\,x^2 + 91$

$= 3x^4 - 4(9)x^3 + 6(3)(6)x^2 + 91$

$= 3x^4 - 36x^3 + 108x^2 + 1$

$\therefore f'(x) = 12x^3 - 108x^2 + 216x = 12x(x^2 - 9x + 18)$

$= 12x(x^2 - 3x - 6x + 18) = 12x(x(x-3) - 6(x-3))$

$= 12x(x-3)(x-6) = 0$

$\therefore$ Stationary Point $\rightarrow$ $x=0, x=3, x=6$

$\Rightarrow f''(x) = 36x^2 - 216x + 216 = 36(x^2 - 6x + 6)$

$f''(0) > 0 \rightarrow$ local minima
$f''(6) > 0 \rightarrow$ local minima
$f''(3) < 0 \rightarrow$ local maxima

$\rightarrow f(0) = 1$ → global minima
$f(3) = 244$ → global maxima
$f(6) = 1$ → global minima

2. $f(x_1, x_2, x_3) = x_1^3 + x_2^3 - 3p^2x_1 - 3q^2x_2 + (x_3 - K)^2 + 91$

$= x_1^3 + x_2^3 - 27x_1 - 27x_2 + (x_3 - 6)^2 + 91$

$\nabla f = 0$

$\Rightarrow \frac{\partial f}{\partial x_1} = 3x_1^2 - 27 = 0 \Rightarrow \boxed{x_1 = \pm 3}$

$\Rightarrow \frac{\partial f}{\partial x_2} = 3x_2^2 - 27 = 0 \Rightarrow \boxed{x_2 = \pm 3}$

$\Rightarrow \frac{\partial f}{\partial x_3} = 2(x_3 - 6) = 0 \Rightarrow \boxed{x_3 = 6}$

Stationary Points
$P_1(3, 3, 6)$
$P_2(3, -3, 6)$
$P_3(-3, 3, 6)$
$P_4(-3, -3, 6)$

$\therefore H(x_1, x_2, x_3) = \begin{bmatrix} 6x_1 & 0 & 0 \\ 0 & 6x_2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

- $H(3,3,6) = \begin{bmatrix} 18 & 0 & 0 \\ 0 & 18 & 0 \\ 0 & 0 & 2 \end{bmatrix} \Rightarrow \begin{matrix} \lambda_1 = 18 \\ \lambda_2 = 18 \\ \lambda_3 = 2 \end{matrix}$ → matrix (+ve) as all $\lambda_m > 0$
  P_1 → local minima.

- $H(3,-3,6) = \begin{bmatrix} 18 & 0 & 0 \\ 0 & -18 & 0 \\ 0 & 0 & 2 \end{bmatrix} \Rightarrow \lambda_m$ → mixed sign → matrix is indefinite
  P_2 → saddle point.

- $H(-3,3,6) \rightarrow P_3$ → saddle point.
- $H(-3,-3,6) \rightarrow \begin{bmatrix} -18 & 0 & 0 \\ 0 & -18 & 0 \\ 0 & 0 & 2 \end{bmatrix} \rightarrow \begin{matrix} \Delta_1 = -18 < 0 \\ \Delta_2 = 324 > 0 \\ \Delta_3 = 648 > 0 \end{matrix}$ → strictly negative
  ~~local Maxima~~ Saddle point.

Key concept → $\lambda_1 * \lambda_2 * \lambda_3$ → does not matter
use $\Delta_1, \Delta_2, \Delta_3$ signs → + + + → +ve definite (minima)
- + - + - → alternate +, - -ve definite
minima
other patter → + - + + → saddle points
↳ (+ve) at beginning always (+ve at start not valid)

5. Minimize $f = 3x_1^2 + 3x_2^2 + x_3^2$
subject to $x_1 + x_2 + x_3 = 16$

$\therefore L(x_1, x_2, x_3, \lambda) = 3x_1^2 + 3x_2^2 + x_3^2 + \lambda(16 - x_1 - x_2 - x_3)$

① $\frac{\partial L}{\partial x_1} = 6x_1 - \lambda = 0 \Rightarrow x_1 = \frac{\lambda}{6}$
② $\frac{\partial L}{\partial x_2} = 6x_2 - \lambda = 0 \Rightarrow x_2 = \frac{\lambda}{6}$
③ $\frac{\partial L}{\partial x_3} = 2x_3 - \lambda = 0 \Rightarrow x_3 = \frac{\lambda}{2}$
④ $\frac{\partial L}{\partial \lambda} = x_1 + x_2 + x_3 = 16$

$\Rightarrow \frac{\lambda}{6} + \frac{\lambda}{6} + \frac{\lambda}{2} = 16$

$\boxed{\lambda^* = 19.2}$

$\therefore x_1^* = 3.2$
$x_2^* = 3.2$
$x_3^* = 9.6$

$\therefore f^* = 3(3.2)^2 + 3(3.2)^2 + (9.6)^2 = 153.6$

$\overline{H} = \begin{bmatrix} 0 & g_1 & g_2 & g_3 \\ g_1 & L_{11} & L_{12} & L_{13} \\ g_2 & L_{21} & L_{22} & L_{23} \\ g_3 & L_{31} & L_{32} & L_{33} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 6 & 0 & 0 \\ 1 & 0 & 6 & 0 \\ 1 & 0 & 0 & 2 \end{bmatrix}$

for $n=3$ variables (3) and $m=1$ constraint, we check the det. $|\bar{H}|$. sign must match $(-1)^{n-m}=-1$ for minimum.

$$|\bar{H}| = -1\begin{vmatrix} 1 & 0 & 0 \\ 1 & 6 & 2 \\ 1 & 0 & 2 \end{vmatrix} + 1\begin{vmatrix} 1 & 6 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 2 \end{vmatrix} + (-1)\begin{vmatrix} 1 & 6 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{vmatrix} = -60$$

$\therefore (-1)^3(-60) = 60 > 0$ the sufficiency condition for a strict local minimum is verified.

4. radius $\to x_1$ $\quad V_0 = 50(3+3) = 300\,m^3$
height $\to x_2$ $\quad$ Minimize $S = 2\pi x_1^2 + 2\pi x_1 x_2$ s.t. $\pi x_1^2 x_2 = V_0 = 300$

$$L(x_1, x_2, \lambda) = 2\pi x_1^2 + 2\pi x_1 x_2 + \lambda(300 - \pi x_1^2 x_2)$$

$$\frac{\partial L}{\partial x_1} = 4\pi x_1 + 2\pi x_2 - 2\pi\lambda x_1 x_2 = 0 \to 2 + \frac{x_2}{x_1} = \lambda x_2 \quad \text{---(1)}$$

$$\frac{\partial L}{\partial x_2} = 2\pi x_1 + (-1)\pi\lambda x_1^2 = 0 \Rightarrow \lambda = \frac{2}{x_1} \quad \text{---(2)}$$

Also $\pi x_1^2 x_2 = 300$

Put $\lambda = \frac{2}{x_1}$ in (1) $\Rightarrow 2 + \frac{x_2}{x_1} = \frac{2}{x_1} x_2 \Rightarrow \boxed{x_2 = 2x_1}$

$\Rightarrow \pi x_1^2(2x_1) = 300 \Rightarrow x_1^3 = \frac{150}{\pi} \approx 47.7465$

$\Rightarrow x_1^* = \left(\frac{150}{\pi}\right)^{1/3} \approx 3.6278\,m$

$\Rightarrow x_2^* = 2x_1 = 7.2557\,m$

$\Rightarrow S^* = 2\pi(3.6278)^2 + 2\pi(3.6278)(7.2557) = 248.1280\,m^2$

$\Rightarrow \lambda^* = \frac{2}{x_1^*} = 0.5513$

$\therefore \Delta S^* \approx \lambda^* \cdot \Delta V_0 \leq 0.5513\,m^2$

5. Minimize $f(x_1, x_2) = (x_1-4)^2 + (x_2-4)^2$ s.t. $x_1 + x_2 \leq 1$, $x_1 \geq 0$, $x_2 \geq 0$

$f(x_1, x_2)$ is a strictly convex quadratic fun. $\{H = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ is pdp definite$\}$

the constraints $g_1(x) = x_1 + x_2 - 1 \leq 0$ $g_2(x) = -x_1 \leq 0$ $g_3(x) = -x_2 \leq 0$ are linear. hence convex. Therefore the problem is convex.

KKT conditions:

① $\nabla f + \mu_1 \nabla g_1 + \mu_2 \nabla g_2 + \mu_3 \nabla g_3 = 0$

$2(x_1 - 4) + \mu_1 - \mu_2 = 0$

$2(x_2 - 4) + \mu_1 - \mu_3 = 0$

② Primal feasibility: $x_1 + x_2 \leq 1$, $x_1 \geq 0$, $x_2 \geq 0$

③ Dual Feasibility: $\mu_1 \geq 0$, $\mu_2 \geq 0$, $\mu_3 \geq 0$

④ complementary slackness: $\mu_1(x_1 + x_2 - 1) = 0$, $\mu_2 x_1 = 0$, $\mu_3 x_2 = 0$

$\therefore$ g_1 active ($\mu_1 > 0$), g_2, g_3 inactive ($\mu_2 = \mu_3 = 0$)

$2(x_1 - 4) + \mu_1 = 0 \Rightarrow x_1 = 4 - \frac{\mu_1}{2}$

$2(x_2 - 4) + \mu_1 = 0 \Rightarrow x_2 = 4 - \frac{\mu_1}{2}$

$\because x_1 + x_2 = 1 \Rightarrow \left(4 - \frac{\mu_1}{2}\right) + \left(4 - \frac{\mu_1}{2}\right) = 1 \Rightarrow 8 - \mu_1 = 1 \Rightarrow \mu_1 = 7 > 0$

Then $x_1^* = 4 - 3.5 = 0.5$, $x_2^* = 4 - 3.5 = 0.5$

Primal Feasibility: $0.5 \geq 0$, $0.5 \geq 0$, $0.5 + 0.5 \leq 1$ (satisfied)

Dual feasibility: $\mu_1 = 7 \geq 0$, $\mu_2 = 0$, $\mu_3 = 0$ (satisfied)

$x^* = (0.5, 0.5)$

$f^* = (0.5 - 4)^2 + (0.5 - 4)^2 = 24.5$

Multipliers: $\mu_1 = 7$, $\mu_2 = 0$, $\mu_3 = 0$

Active constraint: $x_1 + x_2 \leq 1$

6. Maximize $z = 5x_1 + 4x_2$ s.t. $x_1 + 2x_2 \leq 11$, $3x_1 + x_2 \leq 18$, $x_1 \leq 7$

$x_1, x_2 \geq 0$.

$U_1(0,0)$: $z = 0$

$U_2(6,0)$: intersection of $3x_1 + x_2 = 18$ & $x_2 = 0 \Rightarrow z = 5(6) + 0 = 30$

$U_3(5,3)$: intersection of $x_1 + 2x_2 = 11$ & $3x_1 + x_2 = 18$

$3x_1 + 6x_2 = 33 \Rightarrow x_2 = 3$, $x_1 = 5$

$z = 5(5) + 4(3) = 37$

$U_4(0,5.5)$: intersection of $x_1 + 2x_2 = 11$ & $x_1 = 0 \Rightarrow z = 5(0) + 4(5.5) = 22$

constraint $x_1 \leq 7$ & it is redundant because $x_1 \leq 6$ inside feasible region.

$x_1^* = 5$, $x_2^* = 3$ with $z^* = 37$

Binding constraints: $x_1 + 2x_2 \leq 11$ (2) $3x_1 + x_2 \leq 18$

7. Minimize $z = 3x_1 - 3x_2 + 8x_3$ s.t $x_1 + x_2 - x_3 \geq 1$, $2x_1 - x_2 + x_3 \leq 26$

$x_1 + x_2 + x_3 = 9$

$x_1 \geq 0$, $x_2 \leq 0$, $x_3 \rightarrow$ unrestricted.

Let $x_2 = -x_2'$ with $x_2' \geq 0$

$x_3 = x_3^+ - x_3^-$, with $x_3^+, x_3^- \geq 0$

Add surplus $e_1 \geq 0$ to constraint 1 with slack $s_2 \geq 0$ to constraint 2.

Standard form:

Minimize $z = 3x_1 + 3x_2' + 3x_3^+ + 3x_3^-$

s.t $x_1 - x_2' - x_3^+ + x_3^- - e_1 = 1$

$2x_1 + x_2' + x_3^+ - x_3^- + s_2 = 26$

$x_1 - x_2' + x_3^+ - x_3^- = 9$

$x_1, x_2', x_3^+, x_3^-, e_1, s_2 \geq 0$

(b) Basic sol^n:

Maximize $z = 4x_1 + 4x_2$ s.t $2x_1 + x_2 + s_1 = 9$, $x_1 + 3x_2 + s_2 = 15$

$x_1 + 3x_2 + s_2 = 15$, $x_1, x_2, s_1, s_2 \geq 0$

(1) $(x_1, x_2) = (0,0) \Rightarrow s_1 = 9$ & $s_2 = 15$ (set non basic variables $(x_1, x_2) = (0,0)$)

Sol. $(0, 0, 9, 15) \Rightarrow$ Feasible $z = 0$

(2) $(s_1, s_2) = (0,0) \Rightarrow 2x_1 + x_2 = 9$ (1) $x_1 + 3x_2 = 15 \Rightarrow x_1 = 4.2$, $x_2 = 2.4$

(set non basic variable to zero)

Sol. $(2.4, 4.2, 0, 0)$

(3) Set non basic variables $(x_1, s_1) = (0,0)$: Feasible $\Rightarrow$ $z = 26 - 4$

$x_2 = 9$, $3(9) + s_2 = 15 \Rightarrow s_2 = -12$

solution $(0, 9, 0, -12) \rightarrow$ Infeasible (since $s_2 < 0$)

8. Maximize $z = 5x_1 + 6x_2 + 2x_3$

s.t $x_1 + 2x_2 + x_3 + s_1 = 16$

$2x_1 + x_2 + 2x_3 + s_2 = 15$

$x_1 + x_2 + 3x_3 + s_3 = 21$

$x_1, x_2, x_3, s_1, s_2, s_3 \geq 0$

Initial Tableau:

entering variable

Basis	x_1	x_2	x_3	s_1	s_2	s_3	RHS	Ratio Test
s_1 (leaving variable)	1	2 (Pivot)	1	1	0	0	16	→ 16/2 = 8
s_2	2	1	2	0	1	0	15	→ 15/1 = 15
s_3	1	1	3	0	0	1	21	→ 21/1 = 21
z	-5	-6	-2	0	0	0	0	

entering variable

Basis	x_1	x_2	x_3	s_1	s_2	s_3	RHS	Ratio test
x_2	0.5	1	0.5	0.5	0	0	8	→ 8/0.5 = 16
s_2 (leaving variable)	1.5 (Pivot)	0	1.5	-0.5	1	0	7	→ 7/1.5 = 4.6667
s_3	0.5	0	2.5	-0.5	0	1	13	→ 13/0.5 = 26
z	-2	0	1	3	0	0	48	

Basis	x_1	x_2	x_3	s_1	s_2	s_3	RHS
x_2	0	1	0	2/3	-1/3	0	5.6667
x_1	1	0	1	-1/3	2/3	0	4.6667
s_3	0	1	2	-1/3	-1/3	1	10.6667
z	0	0	3	7/3	4/3	0	57.3333

$z_j - c_j \geq 0$

$\therefore x_1^* = 14/3 = 4.6667$

$x_2^* = 17/3 = 5.6667$

$x_3^* = 0$

$Z^* = \frac{172}{3} \approx 57.3333$

Ans

(9) Minimize $z = 4x_1 + 5x_2$

$x_1 + x_2 \geq 5$
$x_1 + 3x_2 \geq 9$
$x_1 + x_2 \leq 18$
$x_1, x_2 \geq 0$

$\longrightarrow$

$x_1 + x_2 - e_1 + A_1 = 5$
$x_1 + 3x_2 - e_2 + A_2 = 0$
$x_1 + x_2 + s_3 = 0$
with $x_1, x_2, e_1, e_2, s_3, A_1, A_2 \geq 0$

Phase I $w = A_1 + A_2$

$w = (5 - x_1 - x_2 + e_1) + (9 - x_1 - 3x_2 + e_2) = 14 - 2x_1 - 4x_2 + e_1 + e_2$

Row 0: $-2x_1 - 4x_2 + e_1 + e_2 = -14$

Tableau I (Initial Phase I):

Basis	x_1	x_2 ← entering	e_1	e_2	s_3	A_1	A_2	RHS
A_1	1	1	-1	0	0	1	0	5 → 5/1
A_2 (leaving)	1	3	0	-1	0	0	1	9 → 9/3 = 3
s_3	1	1	0	0	1	0	0	18 → 18/1 = 18
z	-2	-4	1	1	0	0	0	-14

Tableau 2

Basis	z ← entering	x_2	e_1	e_2	s_3	A_1	A_2	RHS
A_1 (leaving)	2/3	0	-1	1/3	0	1	-1/3	2 → 2/(2/3)
x_2	1/3	1	0	-1/3	0	0	1/3	3 → 3/(1/3)
x_3	2/3	0	0	1/3	1	0	-1/3	15 → 15/(2/3)
z	-2/3	0	1	-1/3	0	0	4/3	-2

Tableau 3

Basis	x_1	x_2	e_1	e_2	s_3	RHS
x_1	1	0	-1.5	0.5	0	3
x_2	0	1	0.5	-0.5	0	2
s_3	0	0	1.0	0	1	13
z	0	0	0	0	0	0

$W^* = 0$, both artificial variables are driven out of the basis

Phase II

Substitute $x_1 = 3 + 1.5e_1 - 0.5e_2$ (4) $x_2 = 2 - 0.5e_1 + 0.5e_2$ in z:

$z = 4x_1 + 5x_2 - 22 + 3.5e_1 + 0.5e_2 \Rightarrow$ row z: $3.5e_1 + 0.5e_2 = -22$

Basis	x_1	x_2	e_1	e_2	S_3	RHS
x_1	1	0	-1.5	0.5	0	3
x_2	0	1	0.5	-0.5	0	2
S_3	0	0	1	0	1	13
z	0	0	3.5	0.5	0	-22

All reduced cost ≥ 0. optimal reached

$x_1^* = 3 \quad x_2^* = 2 \quad z^* = 22$

10. (a) Unbounded: Matrix has a column with all entries ≤ 0 during pivot ratio test (e.g. direction of unbounded ray).

(b) Alternative optima!

Maximize $Z = x_1 + 2x_2$ s.t $x_1 + 2x_2 \leq 13$, $x_1 \leq 6$, $x_2 \leq 8$

objective fcn is parallel to constraints $x_1 + 2x_2 = 13$.

Two optimal vertices: $V_1(0, 6.5)$ & $V_2(6, 3.5)$

$$x(\alpha) = \alpha \begin{bmatrix} 0 \\ 6.5 \end{bmatrix} + (1-\alpha) \begin{bmatrix} 6 \\ 3.5 \end{bmatrix} = \begin{bmatrix} 6(1-\alpha) \\ 3.5 + 3\alpha \end{bmatrix} \rightarrow \alpha \in [0, 1].$$

© Infeasible: Artificial variable staying in the basis with +ve value at the end of Phase I. (constraints $x_1+x_2 \leq 5$ & $x_1+x_2 \geq 6$ are mutually exclusive.

11. Maximize $z = 5x_1 + 4x_2 + 4x_3$ s.t $x_1 + x_2 + x_3 \leq 16$ (y_1)
$2x_1 - x_2 + x_3 \geq 1$ (y_2)
$x_1 + 2x_2 - x_3 = 7$ (y_3)
$x_1, x_2 \geq 0$, x_3 unrestricted

Dual Problem:

Minimize $w = 16y_1 + y_2 + 7y_3$

s.t $y_1 + 2y_2 + y_3 \geq 5$ (Primal $x_1 \geq 0$)
$y_1 - y_2 + 2y_3 \geq 4$ (" $x_2 \geq 0 \Rightarrow$ Dual $\geq$)
$y_1 + y_2 - y_3 = 4$ (" x_3 unrestricted $\Rightarrow$ Dual equality)

Sign Restriction: $y_1 \geq 0$ (Primal $\leq$), $y_2 \leq 0$ (Primal $\geq$), y_3 unrestricted

12. From problem 8

Basic	x_1	x_2	x_3	S_1	S_2	S_3	RHS
z	0	0	3	7/3	4/3	0	172/3

Dual values (y^*) read directly from reduced cost of slack variables (S_1, S_2, S_3)

$y_1^* = 7/3 = 2.3333$

$y_2^* = 4/3 = 1.3333$

$y_3^* = 0$

Dual objective check

$$w^* = b^T y^* = 16\left(\frac{7}{3}\right) + 15\left(\frac{4}{3}\right) + 21(0) = \frac{112}{3} + \frac{60}{3} = \frac{172}{3} = z^*$$

13. Minimize $z = 4x_1 + 4x_2 + 2x_3$

s.t $x_1 + 2x_2 + x_3 \geq 10$, $2x_1 + x_2 + 3x_3 \geq 15$, $x_1 + x_2 + 2x_3 \geq 7$, $x \geq 0$

Multiply constraints by -1 & add slacks S_1, S_2, S_3.

⇒ Starting Tableu:

entering variable

Basis	x_1	x_2	x_3	S_1	S_2	S_3	RHS	
S_1	-1	-2	-1	1	0	0	-10	$\rightarrow 4/\lvert -2\rvert = 2$
leaving → S_2	-2	-1	-3 (pivot)	0	1	0	-15	$\leftarrow 4/\lvert -1\rvert = 4$
S_3	-1	-1	-2	0	0	1	-7	$\leftarrow 2/\lvert -3\rvert = 0.6667$
z	4	4	2	0	0	0	0	

(13) Iteration 1 Tableau:

entering: x_2 (circled)

leaving: s_1 (circled)

Basis	x_1	x_2	x_3	s_1	s_2	s_3	RHS
s_1	-0.3333	-1.6667	0	1	-0.0333	0	-5
x_3	-0.6667	-0.3333	1	0	-0.0333	0	5
s_3	0.3333	-0.3333	0	0	-0.6667	1	3
z	2.6667	3.3333	0	0	0.6667	0	-10

Iteration 2

Basis	x_1	x_2	x_3	s_1	s_2	s_3	RHS
x_2	0.2	1	0	-0.6	0.2	0	3
x_3	0.6	0	1	0.2	-0.4	0	4
s_3	0.4	0	0	-0.2	-0.6	1	4
z	2.0	0	0	2.0	0	0	-20

$\therefore \quad x_1^* = 2 \;,\; x_2^* = 3 \;,\; x_3^* = 4 \;,\; z^* = 20$

14. $f(\lambda) = 2\lambda^2 + \frac{40}{\lambda}$, $\lambda > 0$ (for $p=3 \Rightarrow 16+8(3)=40$)

(a) Unrestricted search with Accelerated step.

Start at $\lambda_0 = 0.5$, initial step $s = 0.1$ doubling each time

1. $\lambda_0 = 0.5 \Rightarrow f(0.5) = 80.5$
2. $\lambda_1 = 0.5 + 0.1 = 0.6 \Rightarrow f(0.6) = 67.3867$ (Decreased)
3. $\lambda_2 = 0.6 + 0.2 = 0.8 \Rightarrow f(0.8) = 51.28$ (Decreased)
4. $\lambda_3 = 0.8 + 0.4 = 1.2 \rightarrow f(1.2) = 36.2133$ (Decreased)
5. $\lambda_4 = 1.2 + 0.8 = 2 \rightarrow f(2) = 28.8$ (Decreased)
6. $\lambda_5 = 2.0 + 1.6 = 3.6 \rightarrow f(3.6) = 37.0811$ (increase)

↳ Bracket: $[1.2, 3.6]$

(b) Exhaustive search on $[0.5, 5.0]$ with $n = 8$ interior points

step size $\Delta x = \frac{5.0 - 0.5}{8+1} = 0.5$

Points: $0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0$.

$\therefore f(1.5) = 31.1667$

$f(2.0) = 28.0000$ (Minimum)

$f(2.5) = 28.50000$

Bracket: $[1.5, 2.5]$

15. $c = 3 + 3 + 0.1(6) = 6.6$

$f(\lambda) = e^{\lambda} - 6.6\lambda$, $\lambda \in [0, 3]$

Iteration 1 $(a = 0, b = 3)$

$m = \frac{3}{2} = 1.5$ (midpoint)

$\lambda_1 = 1.5 - 0.0005 = 1.4995 \Rightarrow f(\lambda_1) = e^{1.4995} - 6.6(1.4995) = -5.4172$

$\lambda_2 = 1.5 + 0.0005 = 1.5005 \Rightarrow f(\lambda_2) = e^{1.5005} - 6.6(1.5005) = -5.4172$

since $f(\lambda_1) > f(\lambda_2)$ discard $[0, 1.4995]$

New interval: $[1.4995, 3.0]$

Iteration 2 $(a = 1.4995, b = 3)$: $m = 2.24975$

$\lambda_1 = 2.24925 \Rightarrow f(\lambda_1) = -5.3643$

$\lambda_2 = 2.25025 \Rightarrow f(\lambda_2) = -5.3614$

$f(\lambda_1) < f(\lambda_2) \Rightarrow$ discard $[2.25025, 3.0]$ ⇒ New interval ⇒ $[1.4995, 2.25025]$

Ans

(16) $c = 1 + 1 + 0.1(6) \approx 2.6$

$$f(\lambda) = \lambda^4 - 10.4\lambda \;,\; \lambda \in [0,3]$$

Fibonacci Method ($n=11$).

Fibonacci sequence: $F_0=1$, $F_1=1$, $F_2=2$, $F_3=3$, $F_4=5$, $F_5=8$

$F_6=13$, F_7 21 . .

initial length: $L_0 = 3-0 = 3$

First modified length $L_2^* = \frac{F_0}{F_{11}} L_0 = \frac{55}{144} \times 3 = 1.1458$

First two experiment placements

Reduction ①

$\lambda_1 = 3 - 1.1458 = 1.8542 \Rightarrow f(1.8542) = -7.446$

$\lambda_2 = 0 + 1.1458 = 1.1458 \Rightarrow f(1.1458) = -10.192$

$f(\lambda_1) > f(\lambda_2) \Rightarrow$ discard $[1.8542, 3.0]$

New interval: $[0, 1.8542]$.

16 Reduction ②

$L_3^* = \frac{F_8}{F_{11}} L_0 = \frac{34}{144}(3) = 0.7083$. Reuse $\lambda_1 = 1.1458$. New point $\lambda_3 = 0 + 0.7083 = 0.7083$ $(f = -7.1149)$

$f(\lambda_1) < f(\lambda_3) \Rightarrow$ discard $[0, 0.7083]$.

Interval: $[0.7083, 1.8543]$ Length $= 1.1458$

Reduction 3:

$L_4^* = \frac{F_7}{F_{11}} L_0 = \frac{21}{144}(3) = 0.4375$. Reuse $\lambda_1 = 1.1458$. New point

$\lambda_4 = 1.8542 - 0.4375 = 1.4167$ $(f = -10.7055)$.

$f(\lambda_4) < f(\lambda_1) \Rightarrow$ discard $[0.7083, 1.1458]$

Interval: $[1.1458, 1.8543]$ Length $= 0.7083$

2. Golden Section Method

Reduction 1: Interval $[0, 3]$

$\lambda_1 = 3(1 - 0.618034) = 1.1459$ $(f = -10.1932)$

$\lambda_2 = 3(0.618034) = 1.8541$ $(f = -7.4649)$

$f(\lambda_1) < f(\lambda_2) \Rightarrow$ discard $[1.8541, 3.0]$. Interval: $[0, 1.8541]$

Length: 1.8541.

Reduction 2: Interval $[0, 1.8541]$. Reuse $\lambda_2' = 1.1459$ $(f = -10.1932)$

New $\lambda_1' = 1.8541(1 - 0.618034) = 0.7082$ $(f = -7.1138)$.

$f(\lambda_2') < f(\lambda_1') \Rightarrow$ discard $[0, 0.7082]$

Interval $[0.7082, 1.8541]$ length $= 1.1459$.

Reduction 3: Interval $[0.7082, 1.8541]$. Reuse $\lambda_1'' = 1.1459$ $(f = -10.1932)$

New $\lambda_2'' = 0.7082 + 0.618034(1.1459) = 1.4164$ $(f = -10.7058)$

$f(\lambda_2'') < f(\lambda_1'') \Rightarrow$ discard $[0.7082, 1.1459]$.

Interval: $[1.1459, 1.8541]$

Length $= 0.7082$

17. $f(\lambda) = \lambda^9 - 6\lambda^2 - 5\lambda + 5$

$A = 0 \quad t_0 = 0.5$

$f(0) = 5$

$\rightarrow f(0.5) = 1.06255 < f(0)$

Double step $t_1 = 1.0 \Rightarrow f(1.0) = -5$

" " $t_2 = 2.0 \Rightarrow f(2.0) = -13$

" " $t_3 = 4.0 \Rightarrow 145 > f(2.0)$

∴ Three points: $A = 1.0$, $B = 2.0$, $C = 4.0$ with $f_A = -5$, $f_B = -13$, $f_C = 145$

Fit parabola $h(\lambda) = a + b\lambda + c\lambda^2$

$$\tilde{\lambda}^* = \frac{f_A(B^2 - C^2) + f_B(C^2 - A^2) + f_C(A^2 - B^2)}{2\left[f_A(B - C) + f_B(C - A) + f_C(A - B)\right]}$$

$$\lambda^* = \frac{-5(4 - 16) - 13(16 - 1) + 145(1 - 4)}{2\left[-5(2 - 4) - 13(4 - 1) + 145(1 - 2)\right]} = 1.0379$$

Ans

(A) 1. Stopping Test Evaluation: with $\bar{\lambda}^* = 1.6379$

$$h(\lambda^*) = -16.8017$$

$$f(\lambda^*) = -12.0891$$

Test: $\left| \frac{h(\bar{\lambda}^*) - f(\bar{\lambda}^*)}{f(\bar{\lambda}^*)} \right| = 0.3898 > 0.001$ (test fails proceed to refit)

Refit 1. because $\bar{\lambda}^* = 1.637946$ (B1) $f(\bar{\lambda}^*) > f_B$ new points are:

$A = 1.6379$ ($f_A = -12.0891$)
$B = 2.0$ ($f_B = -13.0000$)
$C = 4.0$ ($f_C = 145.0000$)

→ Refitted quad. min $\bar{\lambda}^{*''} = 1.8554$

18. $c = 10 + 2(1) + 6 = 18$

$f(\lambda) = \lambda^5 - 5\lambda^3 - 18\lambda + 13$, $f'(\lambda) = 5\lambda^4 - 15\lambda^2 - 18$

$f'(0) = -18 < 0$

$t_0 = 0.4 \Rightarrow f'(0.4) = -20.272 < 0$

$t_1 = 0.8 \Rightarrow f'(0.8) = -25.552 < 0$

$t_2 = 1.6 \Rightarrow f'(1.6) = -23.632 < 0$

$t_3 = 3.2 \Rightarrow f'(3.2) = 352.688 > 0$

→ Bracketing Points:

$A = 1.6$, $B = 3.2$

$f'_A = -23.632$

$f'_B = 352.688$

(18) $A = 1.6 \quad B = 3.2$

$$z = 3\frac{(f_A - f_B)}{B - A} + f'_A + f'_B = 42.3712$$

$$d = \sqrt{z^2 - f'_A \cdot f'_B} = 100.6481$$

$$\bar{\lambda}^* = A + \frac{f'_A + z + d}{f''_A + f''_B + 2z}(B - A) = 2.0616$$

Derivative Test

$$f'(2.0616) = 5(2.0616)^2 - 15(2.0616) - 18 = 0.570770$$

$\therefore f'(\bar{\lambda}^*) > 0$, $\bar{\lambda}^*$ replaced endpoint B.

New bracket $[A, B] = [1.6000, 2.0616]$.

19. Maximize $z = 6x_1 + 5x_2$ s.t $5x_1 + 4x_2 \leq 28$, $2x_1 + 5x_2 \leq 15$, $x_1, x_2 \geq 0$

Intersection of $5x_1 + 4x_2 = 28$ (1) $2x_1 + 5x_2 = 15$

$x_1 = \frac{80}{17} \approx 4.7059$, $x_2 = \frac{19}{17} \approx 1.1176$, $z = \frac{575}{17} \approx 83.8235$.

Branching on $x_1 \leq 4$ (1) $x_1 \geq 5$

child model ($x_1 \leq 4$): Max z occurs at $x_1 = 4$, $z(4) + 5x_2 = 15$

Sol^n: $x_1 = 4$, $x_2 = 1.4$, $z = 31$ $x_2 = 1.4$

child model ($x_1 \geq 5$): from $5(5) + 4(x_2) \leq 28 \Rightarrow x_2 \leq 0.75$

Max z occurs at $x_1 = 5$, $x_2 = 0.75$, $z = 33.75$.

Ans

(19)

$X_1' = 4.7059$, $X_2 = 1.1176$

$Z = 33.8235$

(Branched on X_1)

$(X_1 <= 4)$ → $X_1 = 4$, $X_2 = 1.4$

$Z = 31.0000$

(Branched on X_2)

$X_2 <= 1$ → $X_1 = 4$, $X_2 = 1$

$Z = 29.0000$

(Integer)

(Incubent)

$X_2 >= 2$ → $X_1 = 2.5$, $X_2 = 2$

$Z = 25.0000$

(Fathomed)

(Bound <= 29)

$(X_1 >= 5)$ → $X_1 = 5$, $X_2 = 0.75$

$Z = 33.7500$

(Branched on X_2)

$X_2 <= 0$ → $X_1 = 5.6$ $X_2 = 0$

$Z = 33.6000$

(Branched $X_1 \leq 5$)

$X_1 = 5$, $X_2 = 0$

$Z = 30.0000$

(optimal Incubment)

$X_2 >= 1$ → Infeasible

$X_1^* = 5$, $X_2^* = 0$, $Z^* = 30.0000$

20. Values $v = (13, 16, 8, 14, 11, 9)$.
weight $w = (4, 6, 3, 5, 5, 4)$
capacity $W = 14$

Value-to-weight Ratios:
① Item 1 : 13/4 = 3.25
② " 2 : 16/6 = 2.67
③ " 3 : 8/3 = 2.67
④ " 4 : 14/5 = 2.80
⑤ " 5 : 11/5 = 2.20
⑥ " 6 : 9/4 = 2.25

→ sorted order:
① > ④ > ② > ③ > ⑥ > ⑤

Root Node
Take item 1 (wt 4, val 13) ⇒ Rem cap = 10
" " 4 (wt. 5, val 14) ⇒ Rem cap = 5
" 5/6 of item 2 (wt. 5, val. $16 \times \frac{5}{6} = 13.33$).

Upper Bound $z_{UP} = 13 + 14 + 13.33 = 40.33$. Fractional item: item 2.

(20)

Take item 1, 4 & 5/6 of item 2

$Z_LP = 40.33$

$x_2 = 0$

Capicity = 14

Greedy: item 1, 4, 3, 0.5 of 6

$Z_LP = 39.5000$

$x_6 = 0$

$Lp_2 = 39.40$

$x_6 = 1$

$Z_LP = 38.67$

$x_2 = 1$

Remaining capicity = 8

Greedy: item 1, 0.8 of item 4

$Z_LP = 40.2000$

$x_4 = 0$

$Z_LP = 39.25$

$x_4 = 1$

$Z_LP = 39.75$

$Z^* = 38.0000$ with $X^* = (1, 0, 0, 1, 1, 0)$

Q1. (21.1) Energy per km: $e(v) = c_1 v^2 + \frac{c_2}{v^2}$.

Range: $R(v) = \frac{E}{e(v)} = \frac{E}{c_1 v^2 + c_2 v^{-2}}$

To maximize $R(v)$, we minimize $e(v)$!

$$e'(v) = 2c_1 v - \frac{2c_2}{v^3} = 0 \Rightarrow 2c_1 v = \frac{2c_2}{v^3} \Rightarrow v^4 = \frac{c_2}{c_1} \Rightarrow v_R^* = \left(\frac{c_2}{c_1}\right)^{1/4}$$

Second Derivative Verification:

$$e''(v) = 2c_1 + \frac{6c_2}{v^4}$$

$\because c_1, c_2, v > 0$. $e''(v_R^*) > 0$ strictly, confirming v_R^* minimizes energy per km (21) maximizes range.

(21.2) (1) Formulation (2) LP Relaxation Dimensions.

Number of demand points $n = 40 + 2(6) = 52$

No. of candidate hubs $m = 12$

Variables: m binary variables y_j: $1 \times m = 52 \times 12 = 624$

Continious variables z_{ij}: 636 variables

constraints: $n = 52$ demand constraints + $m = 12$ capicity constraints + $4 \times m = 624$ strong linking constraints

$\Rightarrow$ 688 constraints.

(2) $$\max \sum_{i=1}^{52} u_i \quad \text{s.t.} \begin{cases} u_i - d_i w_j - z_{ij} \le D_{ij} c(v) & \forall i,j \\ Q_j w_j + \sum_{i=1}^{52} z_{ij} \le f_j & \\ w_j \ge 0,\ z_{ij} \ge 0,\ u_i \text{ free} & \end{cases}$$

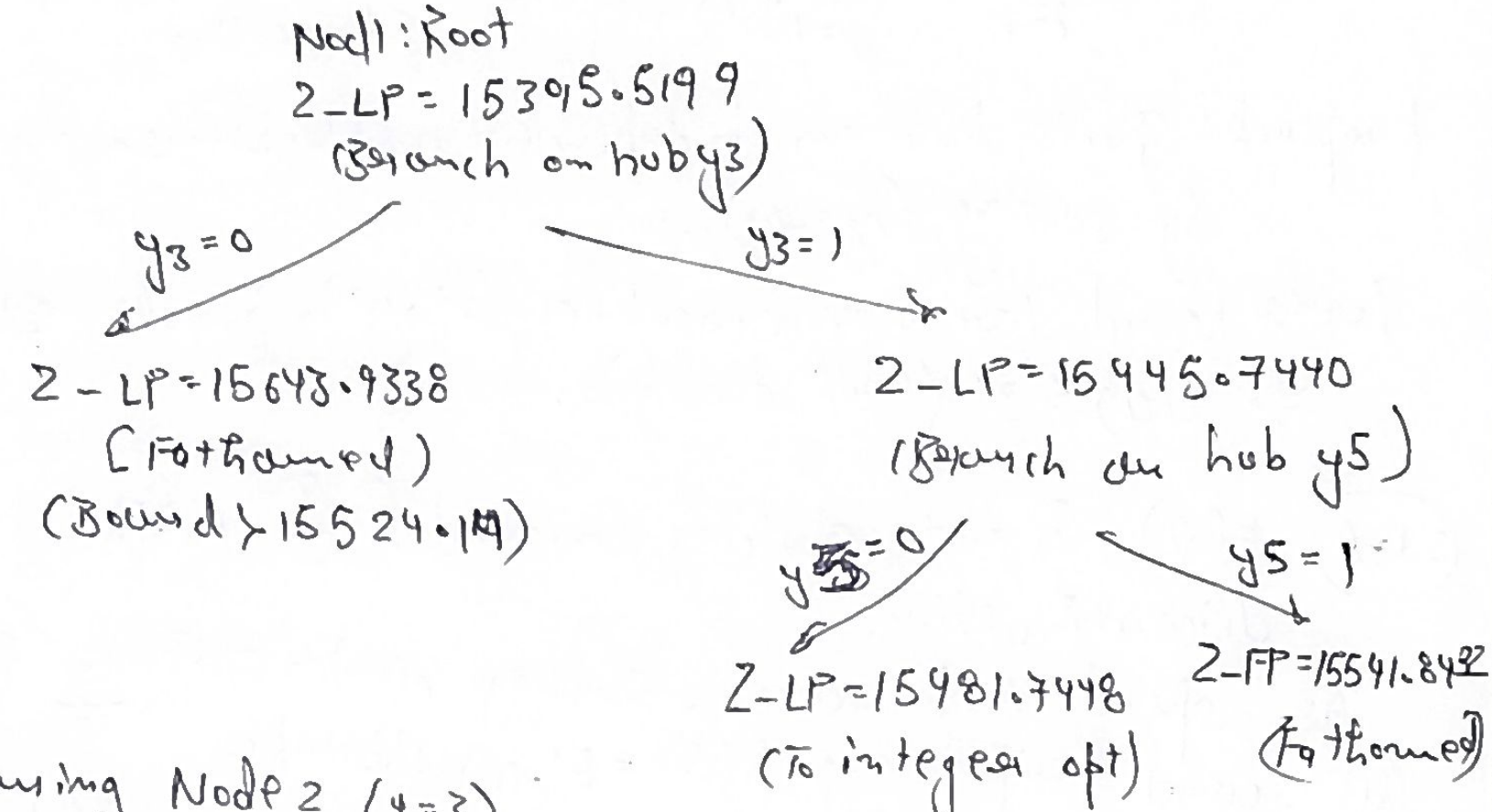
(3) Multiplier Meanings (Text)

u_i: Marginal cost of serving one extra unit of demand from point i

w_j: shadow price of hub j's capicity

(p103) Root LP: $z_{LP} = 15395.5199$

Best Incumbent: Best Bound + Most-Fractional ($z^* = 15524.1936$)



Fathoming Node 2 ($y=3$)

has $z_{LP} = 15643.93 > 15524.19$ (incubment), so it is fathomed by Bound.

(p104) 1-Hub, 2 point Example: $d_1 = 1, d_2 = 1, Q_1 = 2$

- Aggregated: $x_{11} + x_{21} \leq 2y_1 \Rightarrow (1, 0, 0.5)$ is feasible (pays half of fixed cost)
- Strong: $x_{11} \leq y_1 \Rightarrow x_{11} = 1$ forces $y = 1$ (cuts off fractional point).

Bounds: Strong gap = 0.83% ($Z_{LP} = 15395.52$) vs Aggregated gap = 13.28% ($Z_{LP} = 13461.98$).

For 1000 points use the aggregated form + Branch-and-cut (dynamically add violated $x_{ij} \leq y_i$ constraints).

(21-5) Derivation: $\sum_j x_{ij} = 1$ with free multipliers u_i:

$$L(u) = \sum_{i=1}^{n} u_i + \sum_{j=1}^{m} \min_{y_j \in \{0,1\}} \left[f_j y_j + \min_{\sum_i d_i x_i \le D_j y_j} \sum_{i=1}^{n} (c_{ij} - u_i) x_{ij} \right]$$

Knapsack: splits into m independent subproblems. Solved greedily by sorting $(c_{ij} - u_i)/d_i$

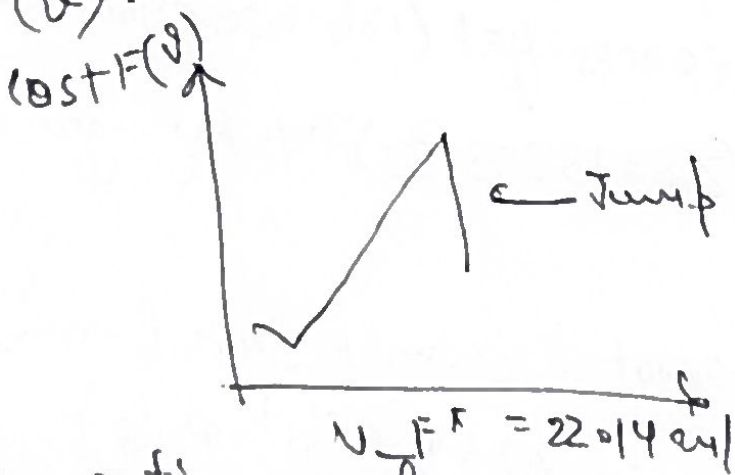
Lower bound: For any feasible (x^*, y^*), penalty $\le \sum_i u_i (1 - \sum_j x_{ij}^*) = 0$

$\Rightarrow L(u) \le z^*$ (weak duality).

(21-6) $F(v)$ is not strictly unimodal (f1) exhibits jump discontinuities.

As speed v increases, energy $e(v) = c_1 v^2 + \frac{c_2}{v}$ increases (f1) round trip range $R(v) = E/e(v)$ shrinks.

Since hub j can serve customers i only $2 D_{ij} \le R(v)$, crossing critical speed thresholds abruptly eliminates pairs (i,j) where $2 D_{ij} > R(v)$. The network must then reassign demand to surviving hubs or open an extra hub (paying fixed cost f_j), causing an instantaneous upward step jump in $F(v)$.



golden section search relies on strict unimodality. Due to the step jumps (f1) multiple local minima, it loses its convergence guarantee (f1) can discard the interval containing the true global jump.

To make search safe:

① Evaluate $F(V)$ on a coarse grid ($V = 8, 9, \ldots 30$ m/s) to locate the true global minimum basin & avoid jump boundaries.

② Apply golden section search locally only within the smooth, jump free interval around the best grid point to reach 0.1 m/s tolerance.

Result: $V_p^* = 22.1414$ m/s which is higher than $V_p^* = 18.9883$ m/s because saving flight time cost outweighs the moderate rise in aerodynamic energy cost ($\propto F(V)$).